From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced), 2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST –IV
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	B
2.	C	B	C
3.	C	B	B
4.	D	C	B
5.	B	A	B
6.	A	C	D
7.	D	B	D
8.	B	B	D
9.	A, B, C, D	A, B	A, C, D
10.	A, C	A, B	A, B, C, D
11.	A, C, D	A, D	A, B
12.	A, C	A, B, C	A, B, D
13.	C	A	B
14.	C	D	B
15.	A	A	C
16.	A	B	B
17.	B	A	C
18.	A	A	A
1.	(A) → (q),(B) → (p, q, s, t), (C) → (p, q, s, t), (D) → (r, s)	A → (p, q) B → (p, q) C → (q) D → (r, t)	(A) → (p), (B) → (p, r) (C) → (r), (D) → (p, r)
2.	(A) → (q, r), (B) → (s), (C)→ (p, s, t), (D) → (p, q, r, t)	A → (p, q, s, t) B → (p, r, s) C → (q, s) D → (q, s, t)	(A) → (q), (B) → (p) (C) → (t), (D) → (r)

Physics

PART – I

1. Particle vel = $\frac{dx}{dt} = x_0\ 2\pi n \cos 2\pi\left(nt - \frac{x}{\lambda}\right)$

max particle vel = $x_0 2\pi n$ putting cos term = 1 wave vel = $\frac{\lambda}{T} = n\lambda$

for $x_0 2\pi n = 4n\lambda$

$\therefore\ \lambda = \frac{x_0\pi}{2}$

2. $\frac{1}{2}mv^2 + mg\frac{1}{\sqrt{3}}\sin 60° = \frac{1}{2}mv_0^2 + mg\times 1\times \sin 30°$

$\Rightarrow$ v_0 = 2 m/s

4. $\omega = \frac{1}{\sqrt{LC_{eq}}} = 100\pi$

$\Rightarrow$ T = 1/50 sec

5. $\sin\theta_C = \mu_2/\mu_1$

$1 - \cos^2\theta_C = \left(\frac{\mu_2}{\mu_1}\right)^2$

$1-(\hat{n}.\hat{p})^2 = \left(\frac{\mu_2}{\mu_1}\right)^2$

$1-\left[\frac{4}{\sqrt{25}}\right]^2 = \left(\frac{\mu_2}{\mu_1}\right)^2 \Rightarrow \frac{\mu_2}{\mu_1} = \frac{3}{5} \Rightarrow \mu_2 = \frac{3\sqrt{3}}{5}$

6. $E_n = -3.4$ eV $E_n\alpha\frac{1}{n^2}$

$\therefore\ E_1 - 13.6$ eV

n = 2

Angular momentum = $\frac{nh}{2\pi} = \frac{2h}{2\pi} = \frac{h}{\pi} = 2.11\times 10^{-34}$ Js

7. $v^2 = \omega^2(A^2 - x^2) = 3\omega^2a^2 \Rightarrow \omega = \frac{v}{\sqrt{3}a} = 1$ rad/sec

$x = 2a\sin(\omega t + \phi)$, $v = 2a\omega\cos(\omega t + \phi)$

At t = 0, x = a & v = +ve

$\frac{1}{2} = \sin\phi$

$\Rightarrow\ \phi = n\pi + (-1)^n\frac{\pi}{6} = \begin{cases}\pi/6\\ 5\pi/6\end{cases}$

v at $\phi = \pi/6$ = + ve and v at $\phi = 5\pi/6$ = –ve

So, $\phi = \pi/6$

$x = 2a\sin(\omega t + \phi) = 2a[\sin\omega t\cos\phi + \cos\omega t\sin\phi]$

$x = a(\sqrt{3}\sin t + \cos t)$

8. For a polytropic process

$$\frac{\Delta U}{\Delta Q} = \frac{\frac{R}{\gamma-1}}{\frac{R}{\gamma-1}+\frac{R}{1-x}} = \frac{1}{1+\frac{\gamma-1}{1-x}}$$

For polyatomic gas, $\gamma = 4/3$

$$\Rightarrow \quad \frac{\Delta U}{\Delta Q} = \frac{1}{1+\frac{1}{3(1-x)}}$$

This is max at x = 3/2

9. Use the concept of capacitor.

10. Area in (v_L-t)graph = $L\Delta I$

$$\Rightarrow \quad 2(i_f - 0) = \frac{1}{2}\times 10 \times 2$$

$$\Rightarrow \quad i_f = 5A$$

13 -15. For long time capacitor gets full charged and charge on each capacitor must be same.

$$\frac{Q}{2}+\frac{Q}{3} = 20$$

$Q = 24\ \mu C$

$$\left[\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2\times 2\times 10^{-6}}} = \frac{10}{2} = 50 \text{ rad/s}\right]$$

$Q(t) = 24 \sin \omega t\ \mu C$

$Q(t) = (24 \sin \omega t)\ \mu C$

$$500t = \frac{\pi}{2}(-1)^n + \pi n$$

$$t = \frac{n\pi}{500} + (-1)^n \frac{\pi}{1000}$$

$$E_{max} = \frac{Q_0^2}{2C} = \frac{24\times 10^{-3}\times 24\times 10^{-6}}{2\times 2\times 10^{-6}} = 144 \times 10^{-6} \text{ Joule} = 0.144 \text{ mJ}$$

16. $Q = Q_{AB} + Q_{BC} + Q_{CA}$

$$Q = -5U_0 + 3U_0 + \frac{10U_0}{3}\ln 2.5$$

17. $W_{AB} = \Delta Q_{AB} - \Delta U_{AB} = -5U_0 - (-3U_0) = -2U_0$

18. Process AB $U\rho$ = constant

$$\frac{P}{\rho} = \frac{RT}{M} \text{ and } U \propto t$$

$$\Rightarrow \quad P = \text{const}$$

Process BC $\rightarrow$isochoric

Process CA $\rightarrow$ isothermal

SECTION - B

2. $I = \frac{1}{12} \times 6m \times 36a^2 + m4a^2 + 2ma^2 = 24a^2m$ for p, q, s & t

$p \Rightarrow \vec{v} = \frac{v_0}{9}\hat{j}$, $24ma^2\vec{\omega} = mv_0 2a\hat{k} + 2mv_0 a\hat{k} \Rightarrow \vec{\omega} = \frac{v_0}{6a}\hat{k}$

$q \Rightarrow \vec{v} = \vec{0}$, $24ma^2\vec{\omega} = 2mv_0 2a\hat{k} + 2mv_0 a\hat{k} \Rightarrow \vec{\omega} = \frac{v_0}{4a}\hat{k}$

$s \Rightarrow \vec{v} = \frac{v_0}{3}(-\hat{j})$, $24ma^2\vec{\omega} = mv_0 2a\hat{k} + 2mv_0 a(-\hat{k}) \Rightarrow \vec{\omega} = \vec{0}$

$t \Rightarrow \vec{v} = \frac{4v_0}{9}(-\hat{j})$, $24ma^2\vec{\omega} = 4mv_0 a\hat{k} + 2mv_0 a(-\hat{k}) \Rightarrow \vec{\omega} = \frac{v_0}{12a}\hat{k}$

$r \Rightarrow \vec{v} = \vec{0}$, $(18\ ma^2 + ma^2 + 2m4a^2)\ \vec{\omega} = 4mv_0 a\hat{k} + 2mv_0 a\hat{k} \Rightarrow \vec{\omega} = \frac{6v_0}{25a}\hat{k}$

Chemistry

PART – II

SECTION - A

1. Due to non availability of d-orbitals, boron is unable to expand its octet. Therefore it cannot extend its covalency more than 4.

3. $BaCO_3 + 2HCl \rightarrow BaCl_2 + H_2O + CO_2$

   $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$

4. $2Ag_2CO_3 \xrightarrow{\Delta} 4\,Ag + 2CO_2 + O_2$

   552 gm $Ag_2CO_3 \rightarrow$ 432 g of Ag

   $\therefore$ 2.76 g $Ag_2CO_3 \rightarrow \frac{432}{552} \times 2.76 = 2.16g$

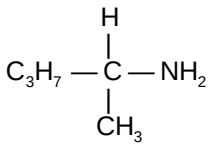
5. NO in iron complex has +1 oxidation number.

   Thus, $a + 5 \times (0) + 1 + 1 \times (-2) = 0$

   $\therefore a = +1$

6. It is called Zinc blend

7. From left to right in a period acidic character increases due to increase in electronegativity.

8. 

9. Roult's law for ideal solutions can be represented in the above two given ways.

11. Hund's rule.

12. Addition of Br_2 in CCl_4 is anti addition.

13. In the original compound

    Z atoms $\rightarrow$ form CCP

    X atoms $\rightarrow$ present in tetrahedral void.

    $\left(\frac{rX}{rZ} = \frac{50}{200} = 0.25\right)$

    Y atoms $\rightarrow$ present in octahedral void

$\left(\frac{rY}{rZ}=\frac{100}{200}=0.5\right)$

Assuming, all voids are occupied, formula of the original compound = $X_8Y_4Z_4$.

On removing atoms along one body diagonal,

We lose → 2Z atoms, 2X atoms, 1Y atom (body centre)

$\therefore$ The new formula becomes → $X_6\ Y_3Z_{3.75}$

Simplest formula → $X_8Y_4Z_5$

14. On removing atoms along another body diagonal, we lose

→ 2Z atoms

→2X atoms

$\therefore$ The new formula becomes = $X_4Y_3Z_{3.5}$

Simplest formula = $X_8Y_6Z_7$

16. $pH = pKa + \log \frac{[Base]}{[Salt]}$

$[Base] = \frac{0.01\times 500}{500} = 0.01$

Let 'a' milli moles of $(NH_4)_2SO_4$

$[NH_4^+] = \frac{a\times 2}{500}$; $\because [Salt] = [NH_4^+]$

$pH = 9.26 + \log\left[\frac{0.01}{2a/500}\right]$

$8.26 = 9.26 + \log\left[\frac{0.01\times 500}{2a}\right]$

a = 25

$\therefore$ Moles of $(NH_4)_2SO_4$ added = 0.025

17. Solution (I) → $\underset{1\ \ 0}{CH_3COONa} + \underset{1\ \ 0}{HCl} \longrightarrow \underset{0\ \ 1}{CH_3COOH} + \underset{0\ \ 1}{NaCl}$

$\therefore \left[CH_3COOH\right] = \frac{\text{No. of moles}}{\text{Volume of solution}} = \frac{1}{1} = 1$

$\left[H^+\right] = C\propto = C\sqrt{\frac{K_a}{C}} = \sqrt{K_a.C} = \sqrt{K_a}$

i.e. $pH_1 = -\frac{1}{2}\log K_a = \frac{1}{2}pK_a$

Solution (II) → $\underset{1}{CH_3COONa} + \underset{1}{CH_3COOH}$

It is buffer.

$\therefore\ pH_2 = pK_a + \log\left[\frac{CH_3COONa}{CH_3COOH}\right]$

$pH_2 = pK_a + \log\frac{1}{1}$

$pH_2 = pK_a$

$\therefore\ \frac{pH_1}{pH_2} = \frac{1}{2}$

18.

	$HA + BOH \rightleftharpoons BA + H_2O$			
$t = 0$	a	12×0.1	0	0
$t = eq$	$(a - 1.2)$	0	1.2	1.2

Meq of HA = Meq of BOH at end point

$A = 26.6 \times 0.1 = 2.66$

$\therefore$ Meq of HA left = 2.66 – 1.2

Now, $pH = pK_a + \log\frac{[Salt]}{[Acid]left}$

$5 = -\log K_a + \log\frac{1.2}{(2.66 - 1.2)}$

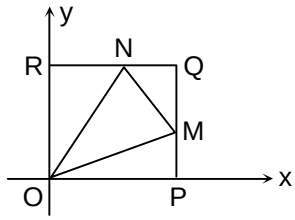
$K_a = 8.219 \times 10^{-6}$

Mathematics

PART – III

SECTION – A

1. We can assume that OP and OR are x–axis and y–axis respectively.
Let OP = a, then ar(sq. OPQR) = a^2



coordinates of M and N are $\left(a, \frac{a}{2}\right)$ and $\left(\frac{a}{2}, a\right)$ respectively

$$\therefore \text{ ar}(\Delta OMN) = \frac{1}{2}\begin{vmatrix} a & \frac{a}{2} \\ \frac{a}{2} & a \end{vmatrix} = \frac{3a^2}{8}$$

$$\therefore \frac{8}{3} = \frac{k}{6}$$

$\therefore$ k = 16

2. Equation of the two circles be $(x - r)^2 + (y - r)^2 = r^2$ i.e.
$x^2 + y^2 - 2rx - 2ry + r^2 = 0$ where r = r_1 and r_2. Condition of orthogonality gives

$$2r_1r_2 + 2r_1r_2 = r_1^2 + r_2^2$$

$$\Rightarrow 4r_1r_2 = r_1^2 + r_2^2$$

Circle passes through (a, b)
$\Rightarrow a^2 + b^2 - 2ra - 2rb + r^2 = 0$ i.e. $r^2 - 2r(a + b) + a^2 + b^2 = 0$
$r_1 + r_2 = 2(a + b)$ and $r_1r_2 = a^2 + b^2$
$\therefore 4(a^2 + b^2) = 4(a + b)^2 - 2(a^2 + b^2)$ i.e. $a^2 - 4ab + b^2 = 0$

3. f(x) is a decreasing function and for major axis to be x–axis
$f(k^2 + 2k + 5) > f(k + 11)$
$\Rightarrow k^2 + 2k + 5 < k + 11$
$\Rightarrow k \in (-3, 2)$

4. $f(x) = 2\sin^2\theta + 4\cos(x + \theta)\sin x \sin\theta + \cos(2x + 2\theta)$
$= 2\sin^2\theta + \cos(2x + 2\theta) + 2\cos(x + \theta)\cos(x - \theta) - 2\cos^2(x + \theta)$
$= 2\sin^2\theta + 2\cos^2(x + \theta) - 1 + 2\cos^2 x - 2\sin^2\theta - 2\cos^2(x + \theta)$
$= \cos 2x$

$$\therefore f^2(x) = f^2\left(\frac{\pi}{4} - x\right) = \cos^2 2x + \sin^2 2x = 1$$

5. Consider the function $f(x) = \frac{a_0x^{n+1}}{n+1} + \frac{a_1x^n}{n} + \frac{a_2x^{n-1}}{n-1} + ... + \frac{a_{n-1}x^2}{2} + a_nx$
Then f(0) = 0 and f(1) = 0
Hence f′(x) = 0 has at least one solution in (0, 1)

6. $fog(x) = \sqrt{e^x - 1}$

$$\therefore I = \int \sqrt{e^x - 1}dx = \int \frac{2t^2}{t^2 + 1}dt \text{ (where } \sqrt{e^x - 1} = t)$$

$$= 2t - 2\tan^{-1} t + C = 2\sqrt{e^x - 1} - 2\tan^{-1}\left(\sqrt{e^x - 1}\right) + C = 2fog(x) - 2\tan^{-1}\left(fog(x)\right) + C$$

$\therefore$ A + B = 2 + (–2) = 0

7. Area $= \int_1^2 (\ell n\, x + \tan^{-1} x)\,dx = x\ell n\, x|_1^2 - \int_1^2 1\,dx + x\tan^{-1}x|_1^2 - \int_1^2 \frac{x}{1+x^2}\,dx$

$= \frac{5}{2}\ell n\, 2 - \frac{1}{2}\ell n\, 5 + 2\tan^{-1}2 - \frac{\pi}{4} - 1$

8. Let $x + y = v$

9. $I + f = (4+\sqrt{15})^n$

Let $g = (4-\sqrt{15})^n$, then $0 < g < 1$

$I + f = {}^nC_0 4^n + {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 + {}^nC_3 4^{n-3}(\sqrt{15})^3 + ...$

$g = {}^nC_0 4^n - {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 - {}^nC_3 4^{n-3}(\sqrt{15})^3 + ...$

$\therefore\ I + f + g = 2\left({}^nC_0 4^n + {}^nC_2 4^{n-2}15 + ...\right)$ = even integer

$\because\ 0 < f + g < 2$

$\Rightarrow f + g = 1$

$\Rightarrow 1 - f = g$ thus I is an odd integer $1 - f = g = (4-\sqrt{15})^n$

$(I + f)(1 - f) = (I + f)\, g = 1$

10. **(A)** $P(E_1) = 1 - P(R\ R\ R) = 1 - \left[\frac{1}{3}\cdot\frac{2}{4}\cdot\frac{3}{5}\right]$

(B) $P(E_2) = 3P(B\ R\ R) = 3\cdot\frac{2}{3}\cdot\frac{1}{4}\cdot\frac{2}{5} = \frac{1}{5}$

(C) $P(E_2) = P(R\ R\ R / R\ R\ R \cup B\ B\ B) = \frac{P(R\ R\ R)}{P(R\ R\ R) + P(B\ B\ B)}$

But $P(B\ B\ B) = \frac{2}{3}\cdot\frac{3}{4}\cdot\frac{4}{5} = \frac{8}{20} \Rightarrow P(E_3) = \frac{0.1}{0.1 + 0.4} = \frac{1}{5}$

(D) $P(E_4) = 1 - P(B\ B\ B) = 1 - \frac{2}{5} = \frac{3}{5}$

11. $|z - i\,Re(z)| = |z - Im(z)|$

Let $z = x + iy$, then $|x + iy - ix| = |x + iy - y|$

i.e. $x^2 + (y - x)^2 = (x - y)^2 + y^2$

i.e. $x^2 = y^2$

i.e. $y = \pm x$

12. $(\lambda - 1)(\vec{a}_1 - \vec{a}_2) + \mu(\vec{a}_2 + \vec{a}_3) + \gamma(\vec{a}_3 + \vec{a}_4 - 2\vec{a}_2) + \vec{a}_3 + \delta\vec{a}_4 = \vec{o}$

i.e. $(\lambda - 1)\vec{a}_1 + (1 - \lambda + \mu - 2\gamma)\vec{a}_2 + (\mu + \gamma + 1)\vec{a}_3 + (\gamma + \delta)\vec{a}_4 = \vec{o}$

Since $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4$ are linearly independent

$\therefore\ \lambda = 1 = 0,\ 1 - \lambda + \mu - 2\gamma = 0,\ \mu + \gamma + 1 = 0,\ \gamma + \delta = 0$

i.e. $\lambda = 1,\ \mu = 2\gamma,\ \mu + \gamma + 1 = 0,\ \gamma + \delta = 0$

i.e. $\lambda = 1,\ \mu = -\frac{2}{3},\ \gamma = -\frac{1}{3},\ \delta = \frac{1}{3}$

13. $\omega_1 = \omega_2 e^{i\frac{4\pi}{3}}$

$\Rightarrow\ \omega_1^3 = \omega_2^3$

$\Rightarrow \omega_1^3\bar{\omega}_1^2\bar{\omega}_2^2 = \omega_2^3\bar{\omega}_1^2\bar{\omega}_2^2$

$\Rightarrow \omega_2\bar{\omega}_1^2 = \omega_1\bar{\omega}_2^2$

14. Since $i\beta$ is real
$\therefore$ β is purely imaginary

15. $-\frac{\alpha}{\alpha} = e^{\pm i\frac{\pi}{2}} = \pm i$

$\therefore (1+i)\left(-\frac{2\theta}{\alpha}\right) = \pm 2(-1+i)$

16. Equation of the second plane is $-x + 2y - 3z + 5 = 0$
$2(-1) + 3,\ 2 + (-4)(-3) > 0$
$\therefore$ origin lies in obtuse angle $(2 \times 1 + 3(-2) - 4 \times 3 + 7)(-1 + 2(-2) - 3 \times 3 + 5)$
$= (2 - 6 - 12 + 7)(-1 - 4 - 9 + 5) > 0$
$\therefore$ P lies in obtuse angle

17. $1 \times 2 + 2 \times 1 - 3 \times 3 < 0$
$\therefore$ Origin lies in acute angle
Also $(2 + 2(-1) - 3(2) + 5)(2 \times 2 - 1 + 3 \times 2 + 1) = (-1)(10) < 0$
$\therefore$ P lies in obtuse angle

18. $1 - 4 - 9 < 0$
$\therefore$ Origin lies in acute angle
Further $(1 + 4 - 6 + 2)(1 - 4 + 6 + 7) > 0$
$\therefore$ The point P lies in acute angle

SECTION – B

1. **(A)** Since $\cos A = \frac{\sin B}{2\sin C}$, we have $\frac{b^2 + c^2 - a^2}{2bc} = \frac{b}{2c}$
or $b^2 + c^2 - a^2 = b^2$ or $c^2 = a^2$
Hence $c = a$ and so the ΔABC is isosceles

(B) $\cos A (\sin B - \sin C) + (\sin 2B - \sin 2C) = 0$
or $\cos A (\sin B - \sin C) + 2 \sin (B - C) \cos (B + C) = 0$
or $\cos A (\sin B - \sin C) - 2 \cos A \sin (B - C) = 0$
$\therefore$ either $\cos A = 0 \Rightarrow A = 90^o$
or $(\sin B - \sin C) - 2(\sin B \cos C - \cos B \sin C) = 0$
$\therefore (b - C) - 2\left[b \cdot \frac{a^2 + b^2 - c^2}{2ab} - c \cdot \frac{c^2 + a^2 - b^2}{2ca}\right]$
or $a(b - c) - 2(b^2 - c^2) = 0$
$\Rightarrow (b - c)[a - 2(b + c)] = 0$
$\therefore b - c = 0$ or $b = c$
$\therefore$ Isosceles

(C) Combine first and thiral and put the value of cos B
$\therefore \frac{2}{ac} \cdot (b) + \frac{1}{b}\frac{c^2 + a^2 - b^2}{2ca} = \frac{a^2 + b^2}{abc}$
or $4b^2 + c^2 + a^2 - b^2 = 2a^2 + 2b^2$
$\therefore b^2 + c^2 + a^2$
$\therefore \angle A = 90^o$

(D) $\dfrac{\sin(A-B)}{\sin(A+B)} = \dfrac{k^2(\sin^2 A - \sin^2 B)}{k^2(\sin^2 A + \sin^2 B)}$ by sine formula

or $\dfrac{\sin(A-B)}{\sin C} = \dfrac{\sin(A-B)\sin(A+B)}{\sin^2 A + \sin^2 B}$

or $\sin(A-B)\left[\dfrac{1}{\sin C} - \dfrac{\sin C}{\sin^2 A + \sin^2 B}\right] = 0$

$\therefore$ Either sin (A – B) = 0

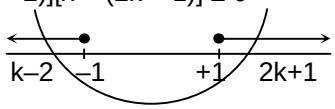
$\therefore$ A = B i.e. Δ is isosceles

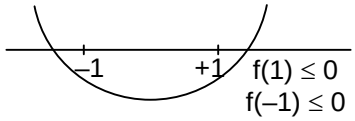
or $\sin^2 A + \sin^2 B = \sin^2 C$ or $a^2 + b^2 = c^2$

$\therefore$ Δ is right angled

2. **(A)** $x^2 - (3k-1)x + 2k^2 - 3k - 2 \geq 0$

$\Rightarrow [x - (k-2)][x - (2k+1)] \geq 0$





Hence $2k + 1 \geq 1$ and $k - 2 \leq -1$

i.e. $k \geq 0$ and $k \leq 1$

$\therefore$ $k \in [0, 1]$

$\therefore$ $\ell + m = 1$

(B) $1 + a \geq 2\sqrt{a}$ (AM $\geq$ GM)

$1 + b \geq 2\sqrt{b}$

$1 + c \geq 2\sqrt{c}$

$1 + d \geq 2\sqrt{d}$

$\therefore$ $(1 + a)(1 + b)(1 + c)(1 + d) \geq 16\sqrt{abcd} = 16$

$\therefore$ Minimum value = 16

(C) $5^{x+2} > 5^{-\frac{2}{x}}$

$\Rightarrow x + 2 > -\dfrac{2}{x}$

$\Rightarrow \dfrac{x^2 + 2x + 2}{x} > 0$

$\Rightarrow \dfrac{(x+1)^2 + 1}{x} > 0$

$\Rightarrow x \in (0, \infty)$

(D) $f(x) = x^3 + 3x + 1$ and g(x) is inverse of f(x)

f(1) = 5

$\therefore$ g(5) = 1

$\therefore$ $g'(f(x)) = \dfrac{1}{f'(x)} = \dfrac{1}{(3x^2 + 3)}$

$\therefore$ $g'(f(1)) = \dfrac{1}{3+3}$

$\Rightarrow g'(5) = \dfrac{1}{6}$